Symmetric wings at a perfect split

Human Compression project: a component for L019

A string C-marking is an ordered family of nonidentity involutions whose nonadjacent entries commute and which satisfies the intersection property for every pair of subsets of its entries. Throughout, groups are actual permutation groups. A group supported on A fixes the complement of A pointwise.

Theorem 1. *Let $\rho_0, \dots, \rho_{r-1}$ be a transitive string C-marking on the disjoint union $\Omega = A \sqcup B$, of size n. Suppose that, for some i, the marks before i are supported on A, the marks after i are supported on B, and*

$$\rho_i = (a\ b)\alpha\beta, \qquad a \in A, \quad b \in B,$$

where α and β are involutions or identities supported on $A \setminus \{a\}$ and $B \setminus \{b\}$, respectively. Suppose also that the omission of ρ_i has exactly the two orbits A, B. Set

$$L = \langle \rho_0, \dots, \rho_{i-1} \rangle, \qquad R = \langle \rho_{i+1}, \dots, \rho_{r-1} \rangle.$$

If $2r > n+2$, then $L = \mathrm{Sym}(A)$ or $R = \mathrm{Sym}(B)$. The empty group on a singleton is understood as its full symmetric group. More precisely, each wing which is not full symmetric has rank at most half its degree.

These are the actual support hypotheses of a perfect split in Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. The theorem applies, in particular, at their threshold $r \geq (n+3)/2$. Its proof does not assert that an arbitrary restriction or block quotient of a C-marking is again a C-marking.

Acquired classification inputs. We use three complete earlier recoveries, with their original attributions and finite premises retained in the dependency register. The transitive rank theorem of Cameron–Fernandes–Leemans–Mixer (L047) classifies every faithful transitive string C-marking of degree d, rank greater than $d/2$, and group different from A_d, S_d. Its exceptions are the odd wreath family in Lemma 3, the complete degree-six/eight imprimitive markings, and the primitive degree-six rank-four S_5 marking. The complete alternating rank theorem of the same four authors (L044) leaves only A_5 of rank three and A_{11} of rank six above half the natural degree. Finally, the acquired alternating census from L071 supplies all four ordered rank-six A_{11} markings, up to simultaneous point conjugacy. Its complete enumeration, including its bound on the searched ranks and its intersection checks, is retained; it is not inferred from a list of examples. We only perform the additional finite tests stated below on these already complete classifications.

Lemma 2. *Let a nontrivial wing in Theorem 1 have degree d and write its marks as $t_0, \dots, t_{k-1}$ in order starting nearest the split. Its attachment point a is moved by t_0 and fixed by all t_j, $j > 0$. If the adjacent split mark is written $q = (a\ b)\alpha\beta$ as above, then*

$$\alpha \in C_{\mathrm{Sym}(A)}(t_1, \dots, t_{k-1})_a.$$

Proof. The omission is the direct product of its two supported wings. Its two stated orbits therefore make each wing transitive on its own support. The actual wing marks inherit the intersection property as a subfamily; their restriction to their support is faithful, so this changes

no group or intersection. For $j > 0$, the string relation gives $t_j q = q t_j$. Since t_j fixes b and $q(b) = a$, it fixes a. Transitivity and $|A| > 1$ then force t_0 to move a. Both $(a\ b)$ and β commute with t_j, so α does too. Its definition already fixes a. □

Call a point moved by the first mark and fixed by every later mark an *endpoint anchor*. The following argument inventories all nonalternating exceptions, including the actual action before any change by an abstract automorphism.

Lemma 3. *No nonalternating proper transitive exception above half the degree in L047 has an endpoint anchor, in either orientation.*

Proof. For the uniform family the degree is $2m$, with $m \geq 5$ odd, and the rank is $m + 1$. The acquired equality proof gives, in the actual two-point block action, the two distinguished base elements λ and $\lambda + e_j$. Here λ flips every block, and $\lambda + e_j$ flips every block except j; the remaining marks act transitively on the blocks. This conclusion precedes the cocycle automorphism used to put their lifts in a convenient abstract normal form.

If a point were moved by just one distinguished mark, that mark would have to be λ. It would also have to lie in block j, since otherwise $\lambda + e_j$ moves it. But some remaining mark moves block j, because their block action is transitive and $m > 1$. It moves both its points, a contradiction. This argument permits all lifts of the block marks and all allowed string orders. The case $m = 3$ is included in the finite census.

There are fourteen retained ordered imprimitive exception tuples: thirteen of degree six and one of degree eight. The primitive exception contributes one further degree-six tuple. For each literal tuple $(t_0, \ldots, t_{k-1})$ the certificate evaluates

$$\{a : t_0(a) \neq a,\ t_j(a) = a \text{ for every } j > 0\}$$

and the same expression for its reverse. All thirty sets are empty. The input is extracted exactly from the complete L047 census, with its indices and hashes retained. Evaluating these sets is an exhaustive point test, and proves the assertion for every tuple in that classification. □

Lemma 4. *Neither A_5 of rank three nor A_{11} of rank six can be a wing in Theorem 1.*

Proof. In A_5, every nonidentity involution has a unique fixed point. For a proposed anchor a of (t_0, t_1, t_2), that point for t_2 would be a. Since t_0 commutes with t_2, it preserves its fixed-point set and fixes a, a contradiction.

For A_{11} the four complete ordered census representatives share the marks

$$t_0 = (8\ 9)(10\ 11), \quad t_1 = (7\ 8)(9\ 10),$$
$$t_3 = (3\ 4)(5\ 6), \qquad t_5 = (1\ 2)(3\ 4),$$

and have the following choices. "First" means the displayed orientation, and "last" means its reverse.

Row	t_2	t_4	First anchors	Last anchors
1	(6 7)(10 11)	(2 3)(4 5)	$\varnothing$	{1}
2	(6 7)(10 11)	(1 3)(2 5)	$\varnothing$	$\varnothing$
3	(6 7)(8 9)	(2 3)(4 5)	{11}	{1}
4	(6 7)(8 9)	(1 3)(2 5)	{11}	$\varnothing$

For each of the four nonempty anchor cases, orient the tuple with the anchor first, and denote it again by $(t_0, \ldots, t_5)$. The complete centralizer fixing its anchor is

$$C_{S_{11}}(t_1, \ldots, t_5)_a = 1. \tag{1}$$

This is checked by an exhaustive equivariant-map calculation: the image of a point determines the images of its whole orbit under the five specified permutations. Start by fixing a; at each still unassigned point try every unused possible image, propagate through all five marks, and reject a conflicting image or repeated value. Every centralizing bijection occurs in exactly one branch. The only surviving bijection in each case is the identity. Thus (1) is a complete centralizer computation, rather than a test of selected involutions. Lemma 2 forces $\alpha = 1$.

Set $u = (a\ 12)$ and consider the first five entries (u, t_0, t_1, t_2, t_3) of the pure attachment. Direct finite closure under right multiplication gives, in every anchor case,

subgroup	order
$U = \langle t_0, t_1, t_2, t_3 \rangle$	5040
$V = \langle u, t_0, t_1, t_2 \rangle$	5040
$U \cap V$	360
$M = \langle t_0, t_1, t_2 \rangle$	60.

(2)

The literal-permutation certificate enumerates these finite closures and intersects the two sets. A separate GAP computation on the exact retained census agrees. In particular $[U \cap V : M] = 6$.

In the actual attachment $q = u\beta$, the permutation β has disjoint support from all five displayed pure marks. The group on the corresponding actual entries is a subgroup of $\langle u, t_0, t_1, t_2, t_3 \rangle \times \langle \beta \rangle$. Projection onto the first factor has central kernel of order at most two. The acquired central-intersection lemma says that, if its marking had the intersection property, the index in (2) would be a power of two: the quotient of the downstairs end intersection by the common-middle image embeds in a quotient of that central kernel. The factor three in six is a contradiction. This also covers $\beta = 1$. As a submarking of a C-marking must satisfy the intersection property, no such wing occurs. □

Proof of Theorem 1. If either support is a singleton, its supported wing is its full symmetric group. Otherwise both wings are faithful transitive string C-groups with the endpoint anchors of Lemma 2. For a proper nonsymmetric wing, the complete L047 theorem and Lemma 3 exclude every nonalternating rank above half the degree. The complete L044 theorem and Lemma 4 exclude every alternating rank above half the degree. Consequently, if both wings were proper, their ranks would satisfy

$$r - 1 = i + (r - i - 1) \leq |A|/2 + |B|/2 = n/2.$$

This contradicts $2r > n + 2$ and proves the theorem. □

Attribution, computation and scope. The three classification theorems remain attributed to their historical authors. Their full acquired proofs and computational premises are dependencies, not new results of this note. The new argument combines their exact exception lists with actual endpoint anchors, the full anchored centralizers, and the central-intersection obstruction. The central-intersection lemma is the separately reviewed component of this L019 recovery. The finite tests here use literal permutation arithmetic independently of GAP; GAP supplies a second check and the earlier complete classifications retain their stated software and library premises. No new census is claimed.

This proves the stated perfect-split wing theorem only. It neither establishes the existence of a perfect split nor completes the extension/contraction bijection or the eventual counting constants of L019. It makes no assertion that every auxiliary proposition in the source is valid, and it grants no whole-original recovery, open-conjecture or Lean credit.